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Observational Study

Diagnostic performance of computed tomography-guided percutaneous bone biopsy in musculoskeletal oncology: A single-center cohort study

Andreea Minodora Stuparu Adam, Alnuaimi Osama, Alexandru Stere, Daniel Cochior, Mark Eduard Pogărăşteanu, Cosmin Alec Moldovan, Ioan-Sorin Tudorache, Razvan Adam, Adina-Diana Moldovan

Andreea Minodora Stuparu Adam, Razvan Adam, Faculty of General Medicine, George Emil Palade University of Medicine, Pharmacy, Science and Technology, Targu-Mures 540142, Mureş, Romania

Alnuaimi Osama, Department of Radiology Imaging, Affidea Clinic, Bucharest 013713, Bucureşti, Romania

Alexandru Stere, Faculty of Materials Science and Engineering, National University of Science and Technology Politehnica Bucharest, Bucharest 060042, Bucureşti, Romania

Daniel Cochior, Cosmin Alec Moldovan, Department of Medical-Surgical and Prophylactic Disciplines, Faculty of Medicine, Titu Maiorescu University, Bucharest 031593, Bucureşti, Romania

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Daniel Cochior, Department of General Surgery, Monza Hospital, Bucharest 021967, Bucureşti, Romania

Mark Eduard Pogărăşteanu, Faculty of Medicine, Carol Davila University of Medicine and Pharmacy, Bucharest 050474, Bucureşti, Romania

Mark Eduard Pogărăşteanu, Department of Orthopaedics and Trauma Surgery, Dr. Carol Davila Central Military Emergency University Hospital, Bucharest 010242, Bucureşti, Romania

Cosmin Alec Moldovan, Department of General Surgery, Witting Clinical Hospital, Bucharest 010243, Bucureşti, Romania

Ioan-Sorin Tudorache, Razvan Adam, Adina-Diana Moldovan, Department of Preclinical Sciences, Faculty of Medicine, Titu Maiorescu University, Bucharest 031593, Bucureşti, Romania

Razvan Adam, Department of Orthopedics, Elias Clinical University Emergency Hospital, Bucharest 011461, Bucureşti, Romania

Adina-Diana Moldovan, Doctoral School in Medicine, Titu Maiorescu University, Bucharest 031593, Bucureşti, Romania

Author contributions: Stuparu Adam AM and Adam R contributed to conceptualization; Stuparu Adam AM and Moldovan CA contributed to methodology; Stere A, Osama A, Pogărăşteanu ME, and Cochior D contributed to validation; Cochior

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D and Moldovan AD contributed to formal analysis; Adam R, Tudorache IS, and Pogărăşteanu ME contributed to investigation; Osama A, Moldovan CA, and Moldovan AD contributed to data curation; Stuparu Adam AM, Adam R, and Stere A contributed to writing-original draft preparation; Cochior D and Moldovan CA contributed to writing-review and editing; Adam R contributed to supervision, project administration.

Corresponding author: Cosmin Alec Moldovan, MD, PhD, Associate Professor, Chief Physician, Department of Medical-Surgical and Prophylactic Disciplines, Faculty of Medicine, Titu Maiorescu University, Gheorghe Petrascu 67A Street, 3rd District, Bucharest 031593, Bucureşti, Romania. cosmin.moldovan@prof.utm.ro

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Abstract

BACKGROUND

Computed tomography guided percutaneous bone biopsy provides tissue for the assessment of suspected primary and secondary bone lesions. Diagnostic yield, concordance with subsequent diagnostic assessment, and adverse events describe different aspects of performance. Their interpretation depends on case selection, outcome definitions, and the completeness and independence of diagnostic verification.

AIM

To describe diagnostic yield, recorded diagnostic concordance, and complications in a retrospective single-center cohort of image-guided bone biopsies.

METHODS

We retrospectively reviewed 64 consecutive image-guided percutaneous bone biopsy procedures performed from January 2023 through December 2025. Diagnostic yield required clinically useful histopathology without repeat sampling. Recorded concordance used a composite of surgical histopathology, immunohistochemistry-supported assessment, or clinical and imaging follow-up. Non-diagnostic procedures remained in the full-cohort analysis. Complications were classified using the legacy Society of Interventional Radiology system. Proportions were summarized with exact binomial 95% confidence intervals (CI); no external hypothesis tests were performed.

RESULTS

Diagnostic yield was 60/64 (93.75%; 95%CI: 84.8%-98.3%). Recorded concordance among 60 diagnostic, evaluable procedures was 60/60 (100%; 95%CI: 94.0%-100%). The full-cohort concordant diagnostic-result rate was 60/64 (93.75%) when four non-diagnostic procedures were counted as failures. Six biopsies revised the pre-procedural

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clinical diagnosis. Pelvic lesions were the most frequent sites. Two minor class B biopsy-tract fistulas occurred (2/64, 3.1%; 95%CI: 0.4%-10.8%); no major complications were recorded (0/64; 95%CI: 0%-5.6%). The heterogeneous cohort, incomplete verification reporting, and potentially non-independent composite reference standard limit interpretation of the conditional concordance estimate.

CONCLUSION

This cohort showed high diagnostic yield and two recorded minor complications. Conditional concordance does not establish unbiased diagnostic accuracy; independent verification and prospective confirmation are needed.

Key Words: Biopsy, Needle; Tomography, X-ray computed; Bone neoplasms; Diagnostic yield; Retrospective studies; Complications

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Core Tip: In 64 consecutive image-guided bone biopsies performed over 36 months, diagnostic yield was 60/64 (93.75%). Recorded concordance was 60/60 among diagnostic, evaluable procedures; counting four non-diagnostic procedures as failures gave 60/64 for the full cohort. Six biopsies corrected the pre-procedural clinical diagnosis. Two minor class B complications were managed conservatively. These are descriptive local results, not evidence of unbiased 100% diagnostic accuracy: verification was incomplete and used a potentially non-independent composite

standard. No predictors were tested and no pooled literature comparison was performed.

INTRODUCTION

Imaging-guided bone biopsies using computed tomography (CT) scans are an important diagnostic method for primary or secondary bone lesions. Although open excisional biopsy is often regarded as the reference standard, CT-guided biopsy may be the only practical diagnostic option in small lesions, lesions measuring 1 cm or less, or intraosseous lesions-most commonly in the pelvis-that are difficult to approach using the classic open method[1-3]. Altered clinical status in oncologic patients, including coagulation disorders or major organ dysfunction, may further limit the feasibility of surgery or general anesthesia[1].

At the same time, CT-guided percutaneous bone biopsies have a low complication rate compared with open biopsy and offer additional practical advantages, including lower morbidity, reduced costs, shorter hospital stay, no use of the operating theater, lower consumption of supplies and instruments, and the possibility of performing the procedure under local anesthesia or intravenous sedation[1,3].

As with open biopsies, percutaneous needle biopsies carry the risk of tumor cell contamination along the biopsy tract. The optimal tract is the shortest route between the skin and the lesion while avoiding neurovascular structures, joints, and vital organs. Because the biopsy tract should be excised *en bloc* with the tumor in cases undergoing curative surgery, it should be planned, as in open biopsy, along the future surgical approach. In metastatic lesions or tumors not intended for curative excision, the shortest and least complex route is preferred. In patients with multiple lesions, the lesion best suited for biopsy should be selected to maximize diagnostic yield while minimizing procedural risk[1].

To increase the likelihood of obtaining diagnostic results, it is recommended to target

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the region of the lesion with the most aggressive imaging appearance, to avoid areas with necrotic appearance when possible, and to collect multiple samples when the lesion is larger than 1 cm in diameter[1,4,5]. In cystic lesions or massive osteolysis with extensive necrosis, multiple samples should be taken from the tumor margins, where viable tumor cells are more likely to be preserved[1]. In addition, bloody fluid aspirated from osteolytic tumors may be sent for cytological pathological examination, and when infectious lesions are part of the differential diagnosis, bacteriological examination should be requested in parallel[1].

One of the possible disadvantages of CT-guided needle biopsies is the collection of small tissue fragments, which may provide limited material for histopathological examination. To improve diagnostic yield, multiple samples should be collected from different tumor areas. Omura *et al*[5] recommended obtaining three samples in bone lesions and four in soft-tissue tumors, whereas Espinosa *et al*[1] recommended obtaining two to three samples when a 14-16-gauge biopsy needle is used.

Good imaging evaluation and identification of the best target lesion are key elements of percutaneous biopsy. In early, metabolically active lesions that have not yet produced major structural bone changes, conventional CT may fail to depict the target adequately. In such cases, functional imaging guidance using positron emission tomography-CT (PET-CT) may improve biopsy efficiency by identifying metabolically active tumor regions and directing the needle toward the most viable part of the lesion[6,7]. In patients with multiple lesions, PET-CT-guided biopsy may also help identify the most active target and reduce sampling error. However, this method requires radioactive tracers and incurs higher costs, so it is best reserved for selected cases[6,7].

Diagnostic yield describes whether sampling produces a clinically useful tissue diagnosis, whereas diagnostic accuracy requires comparison with an adequately established final diagnosis. A change from the pre-biopsy clinical impression is not, by

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itself, evidence of biopsy error. Published musculoskeletal studies use different definitions and verification pathways, so these outcomes should not be treated as interchangeable[2,4,5,8].

The present study describes diagnostic yield, recorded concordance with subsequent diagnostic assessment, and complications in a consecutive single-center series. Its contribution is a description of local practice and individual diagnostic difficulties, rather than a test of superiority, identification of independent predictors, or a change to established diagnostic pathways.

MATERIALS AND METHODS

Study design and setting

This retrospective single-center cohort included 64 image-guided percutaneous bone biopsy procedures performed from January 1, 2023 to December 31, 2025. Procedures were performed by an orthopedic surgeon with oncologic certification and a radiologist with CT and PET-CT competency. The procedure is the unit used for the aggregate analyses reported below. Reporting was revised with reference to the STROBE recommendations[9].

Data were extracted retrospectively from routine institutional clinical, imaging, procedural, and pathology records. There was no prospective allocation, study-specific intervention, or study-mandated follow-up. The procedure-enrollment interval is distinct from the outcome-ascertainment period; the final follow-up date must be specified separately.

The Ethics Committee of Titu Maiorescu University of Bucharest approved the retrospective research protocol on March 23, 2026 (approval No. 7/23.03.2026). The manuscript records that research extraction and analysis began after this approval, whereas the procedures had already been performed as routine care. Written consent for the clinical intervention was obtained at the time of the procedure, and records were

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pseudonymized before analysis. Consent for treatment, authorization for retrospective research, and permission to publish clinical or histopathology images are components of the Patient's Informed Consent form attached to all clinical charts and all patients signed these consents as part of the admission for treatment.

The cohort is a consecutive one and comprises percutaneous CT- or PET-CT-guided sampling of suspected primary or secondary bone lesions during the stated interval; four procedures were performed with radiofrequency ablation. Non-diagnostic biopsy results were retained in the diagnostic-yield denominator.

No a priori sample-size calculation or post-hoc power analysis was performed. The sample size was determined by the consecutive procedures reported during the study interval. Exact confidence intervals describe precision; neither the cohort size nor the observed yield establishes adequate power for predictor analyses or precise safety estimation.

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Eligibility, unit of analysis, and procedural categories

The analysis includes 64 procedure records, with sex recorded as female in 44 and male in 20. The procedural categories comprise 58 conventional CT-guided biopsies, two PET-CT-guided biopsies, and four biopsies associated with CT-guided radiofrequency ablation.

Pre-procedural assessment and target selection

At presentation, patients had either a referral diagnosis or a presumptive diagnosis established after clinical evaluation. The indication for CT-guided biopsy was determined based on lesion location, size, imaging characteristics, and patient clinical status. Prior imaging examinations (CT and/or magnetic resonance imaging) were reviewed jointly with the imaging specialist to select a target region with the highest expected diagnostic yield, while avoiding necrotic-appearing areas when possible. A

minimum laboratory panel was obtained before biopsy, including complete blood count, coagulation tests, and renal and hepatic function tests.

Anesthesia and sedation

Local anesthesia with 1% lidocaine was used in 14 cases. In 50 cases, local anesthesia was combined with intravenous sedation using midazolam and fentanyl. Sedation was preferentially used for deep lesions (typically > 4 cm from the skin surface) and for anxious patients.

Biopsy technique and post-procedural imaging

CT-guided biopsy procedures were performed using a revolution frontier CT scanner (General Electric/GE Healthcare, Chicago, IL, United States). A low-dose planning CT scan (working parameters: 35 mA, 120 kV) was first acquired over a wide field to localize the target lesion. A skin entry point was determined using the laser reference line and a skin-marking method aided by equidistant pins/grid alignment; the scan range was then reduced to the lesion level to confirm the planned trajectory. The position, direction, and depth of the trajectory were verified by CT, initially using the needle as a guide to minimize soft-tissue injury, followed by advancement and tissue sampling under CT guidance.

For lesions > 1 cm, three tissue samples were typically obtained for histopathology; for lesions ≤ 1 cm, one to two samples were collected, limited by lesion size. A Medex MED-L biopsy needle (8 G, 150 mm) was used. At the end of each procedure, a targeted control CT scan was performed to evaluate immediate local complications, including vascular, osseous, or adjacent organ injury.

Outcome definitions

The primary descriptive endpoint was diagnostic yield: The number of procedures

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classified as providing a clinically useful histopathological diagnosis without repeat sampling divided by all 64 included procedures. A result described as uncertain or non-diagnostic was retained as a failure of yield; subsequent refusal of repeat biopsy did not cause the initial sampling failure. The precise case-level criteria for a clinically useful diagnosis require confirmation from the pathology records.

Recorded diagnostic concordance was a secondary, conditional endpoint: Agreement between the biopsy diagnosis and the recorded final diagnostic assessment among procedures with both a diagnostic sample and an available final assessment. The composite assessment comprised surgical histopathology, immunohistochemistry-supported oncologic work-up, or clinical and imaging follow-up. The reported 60/60 estimate is therefore conditional concordance, not independently validated diagnostic accuracy. A full-cohort concordant diagnostic-result rate was also calculated as 60/64, counting all four non-diagnostic procedures as failures. These four procedures were not assigned a benign final diagnosis or labeled false-negative without verification. Six biopsies that differed from the pre-procedural impression but agreed with the recorded final diagnosis were not classified as biopsy errors.

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Complications

Complications were recorded from clinical findings and immediate post-procedural CT. Events were described using the legacy outcome-based Society of Interventional Radiology A-F classification reproduced in Table 1[10]. No retrospective conversion to a newer SIR severity scale was performed. The duration and completeness of surveillance for delayed adverse events were not specified in the aggregate record.

Use of published literature

Published studies were used as a non-systematic, non-exhaustive source of clinical and methodological context. Relevant examples include large musculoskeletal and osseous

biopsy cohorts, a 13-G sampling study, a spinal-biopsy meta-analysis, and a technical review[11-15]. No PRISMA-based study selection, formal risk-of-bias assessment, pooled estimate, or inferential comparison with the present cohort is claimed. Primary studies, reviews, and meta-analyses are identified separately, and their endpoints are not combined.

Statistical analysis

Categorical variables were summarized as counts with explicit denominators. During this revision, aggregate proportions and two-sided exact Clopper-Pearson 95% confidence intervals (CI) were independently recalculated using Python 3.13.5 and SciPy 1.17.0 (scipy.stats.binomtest, proportion_ci with method="exact"). Diagnostic yield and the full-cohort concordant diagnostic-result rate used all 64 procedures; conditional concordance used the 60 evaluable diagnostic procedures; recorded complication rates used 64 procedures. These intervals assume independent observations, pending confirmation of patient-procedure linkage. No subgroup hypothesis tests, adjusted predictor analyses, imputation of final diagnoses, or tests against external literature proportions were performed.

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RESULTS

Cohort characteristics

Sixty-four procedure records were included. Sex was recorded as female in 44/64 (68.8%) and male in 20/64 (31.3%). There were 58 conventional CT-guided biopsies, two PET-CT-guided biopsies, and four biopsies associated with radiofrequency ablation. Local anesthesia alone was used in 14/64 procedures and additional intravenous sedation in 50/64. Age distribution and procedure-level lesion dimensions, depth, and actual core counts were not available in the supplied summary.

The reported spectrum included metastases, primary or recurrent neoplasms, benign

lesions, and non-neoplastic mimics (Table 2). The grouped table lists 10 prostate, 10 breast, and eight other metastatic lesions; these category counts do not support the previously stated metastatic proportion of 59%. That percentage has been withdrawn pending reconciliation with the individual records. Table 2 retains the source category counts transparently and does not provide a verified mutually exclusive breakdown of final verification methods.

In terms of anatomical distribution, lesions were located predominantly in the pelvic bones, especially the iliac wing and iliac body, followed by the femur, humerus, scapula, sternum, calcaneus, peroneal malleolus, and vertebral posterior elements (Figure 1).

Imaging appearance was osteolytic in 28/64 procedures (43.8%), osteosclerotic in 18/64 (28.1%), and mixed in 18/64 (28.1%) (Figure 2).

Procedure disposition and diagnostic-verification availability are shown in Figure 3. The figure distinguishes the 64 included records, 60 diagnostic/evaluable procedures, and four non-diagnostic procedures. Tables 3 and 4 report the endpoint denominators and the verification information available from the aggregate record.

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Diagnostic yield and recorded concordance

Biopsy was diagnostic in 60/64 procedures (93.75%; exact 95%CI: 84.8%-98.3%). Four procedures were non-diagnostic (4/64, 6.25%; exact 95%CI: 1.7%-15.2%). Two involved sclerotic iliac lesions in patients with breast carcinoma. Two other non-diagnostic procedures involved extensive osteolytic iliac lesions in patients with lung carcinoma who declined repeat sampling. Refusal prevented further tissue confirmation rather than explaining why the initial sample was non-diagnostic.

Six procedures yielded diagnoses that differed from the pre-procedural clinical impression and agreed with the recorded final assessment (6/64 procedures; 6/60 diagnostic procedures). They are counted within, not in addition to, the 60 diagnostic

and concordant results.

Among the 60 diagnostic procedures with a recorded final assessment, conditional concordance was 60/60 (100%; exact 95%CI: 94.0%-100%). Counting all four non-diagnostic procedures as unsuccessful outcomes gave a full-cohort concordant diagnostic-result rate of 60/64 (93.75%; exact 95%CI: 84.8%-98.3%). Neither estimate establishes sensitivity, specificity, or accuracy against an independently applied reference standard (Table 3).

Recorded verification included surgical histopathology, immunohistochemistry-supported assessment, and clinical or imaging follow-up. The source table provides grouped descriptions rather than an exclusive count for each verification route; these reporting gaps are displayed in Table 4. Final diagnostic status was not documented for the four non-diagnostic procedures.

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Complications

Two minor biopsy-tract fistulas were recorded (2/64, 3.1%; exact 95%CI: 0.4%-10.8%), both classified as legacy SIR class B. In the patient diagnosed with bone tuberculosis, purulent drainage was treated with local wound care and antibiotics for approximately three weeks. A second fistula had serous drainage and was managed with local care for one week without antibiotics. No major complications were recorded (0/64; exact 95%CI: 0%-5.6%); 62/64 procedures had no recorded complication (96.9%; exact 95%CI: 89.2%-99.6%). All patients have been discharged on the day of the procedure.

Illustrative cases

Case 1: Osteocondensing lesions disseminated in the pelvic area, in a patient with a medical history of prostate carcinoma (Figure 4). The final histopathological result was a secondary bone determination of G3 poorly differentiated adenocarcinoma (Figure 5).

Case 2: Osteolytic bone lesion of the right iliac body, in a patient with a medical history of breast carcinoma (Figure 6). The final histopathological result was a secondary bone determination of poorly differentiated carcinoma (Figure 7).

Case 3: A metabolically active bone lesion of the iliac wing, visible only on PET-CT, in a patient with a medical history of breast carcinoma (Figure 8). The final histopathological result showed bone lamellae and hematopoietic marrow without notable histopathological changes (non-diagnostic result).

DISCUSSION

This single-center series provided diagnostically useful tissue in 60 of 64 procedures. Two minor complications were recorded. The 60/60 concordance result applies only to diagnostic procedures with an available composite final assessment and must not be interpreted as proof of perfect or independently established diagnostic accuracy. The full-cohort concordant diagnostic-result rate is 60/64 and keeps the four sampling failures visible.

Selected published studies and evidence syntheses are summarized in Table 5 as descriptive context[16-18]. They differ in population, guidance method, repeat-biopsy handling, and the endpoint being measured. Adequacy, initial diagnostic yield, sensitivity, diagnostic accuracy, and molecular-testing success are therefore reported separately. No literature average, significance test, equivalence, or non-inferiority conclusion is derived from this selection[19-23].

The metastatic and primary-tumor cases illustrate the clinical need for tissue characterization, but the exact metastatic proportion requires reconciliation of the source records[24-26]. Molecular utility is a distinct outcome that was not measured systematically in this cohort. In a selected series of 50 lytic bone lesions in patients suspected of lung cancer, Toffart *et al*[24] reported pathological diagnostic yield of 50/50

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and molecular-testing success in 35/37 tested specimens.

Lesion characteristics have been associated with biopsy outcomes in other datasets. Li *et al*[21] reported higher yield in lytic than sclerotic lesions and in lesions at least 3 cm compared with smaller lesions. Hwang *et al*[22] described the relationship of cortical destruction and extraosseous components to diagnostic rather than indeterminate results. The meta-analysis by Suh and Yun[23] addressed sclerotic lesions specifically and reported pooled yield of 74% and accuracy of 87%, based on different contributing datasets.

Differences in verification and sampling endpoints are particularly important when reading large series. Monfardini *et al*[25] reported adequate specimens in 290/308 procedures and sensitivity of 96.7% using their own follow-up framework. In the Ferreira *et al*[20] meta-analysis, published in 2023, pooled core-needle diagnostic yield and accuracy were 91.4% and 92.7%, respectively; the combined core-needle/fine-needle approach had pooled yield of 96.5%. These published estimates are not matched control data and were not statistically compared with this cohort.

The Dablan *et al*[17] study is relevant as it includes 226 biopsies in 211 patients, used both ultrasound and CT guidance, and reported adequacy in 148/211 patients (70.1%). Its PET-derived standardized uptake value analysis was performed in the PET-imaged subset. The numerical adequacy percentage must therefore not be labeled as a proportion of 226 biopsies or treated as a CT-only performance estimate.

The observed complication proportion has considerable uncertainty: Its exact 95%CI extends from 0.4% to 10.8%. The absence of recorded major events in 64 procedures is compatible with an underlying major-event proportion up to 5.6% by the two-sided exact interval, before considering incomplete surveillance. Published reports also use different definitions of adverse events; for example, Maciel *et al*[19] reported three complications among 186 procedures, including events that required intervention.

Only four procedures were non-diagnostic. Interval chemotherapy is explicitly

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documented in one detailed case, whereas the treatment history of the second sclerotic failure is not specified. The six diagnoses that revised the clinical impression should not be mislabeled as biopsy errors.

The infectious case should not be benchmarked against oncological diagnostic-accuracy estimates. Hoang *et al*[26] reported positive cultures in 24/115 biopsies undertaken for suspected osteomyelitis. Culture positivity is not the same outcome as histopathological yield or final diagnosis of malignancy, and that study is not specific evidence about tuberculosis-related fistulas.

Only two procedures used PET-CT guidance, and at least one was non-diagnostic. This small subgroup does not permit comparison of CT with PET-CT guidance[6,7].

This study has several limitations. Referral to a specialized single center and the predominance of pelvic targets may select a case mix unlike that of other institutions. The cohort is small and heterogeneous, including benign, malignant, infectious, and ablation-associated procedures. The screening log, handling of repeat or aborted procedures, and patient-procedure linkage must be confirmed before selection completeness and independence of observations can be established.

Partial and differential verification are important limitations. The four non-diagnostic procedures have no documented final diagnosis, and the remaining procedures were assessed using different routes rather than one independent reference standard. Excluding the four failures from conditional concordance can inflate the apparent success rate.

Incorporation and review biases may also inflate concordance. Immunohistochemistry on the index-biopsy sample is part of the diagnostic work-up rather than an independent reference; subsequent treatment and follow-up interpretation may have been informed by that result. Blinding, adjudication rules, and the distribution and duration of verification were not documented in the aggregate record. Accordingly, 100% conditional concordance cannot establish sensitivity,

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specificity, or generalizable diagnostic accuracy.

The literature component is explicitly narrative and non-exhaustive. Its purpose is to distinguish clinically relevant published endpoints and limitations, not to present a systematic evidence base or a pooled comparator.

Future prospective, multicenter work should enumerate all potentially eligible procedures, define handling of repeat sampling, obtain independent final verification where feasible, and document adverse-event follow-up. Predefined lesion morphology, size, depth, actual core number, treatment exposure, and operator variables could then be evaluated with suitable denominators and a sample size chosen for the intended precision or comparison.

CONCLUSION

In this heterogeneous single-center cohort, diagnostic yield was 60/64 and two minor complications were recorded. The 60/60 conditional concordance estimate is limited by selective and potentially non-independent verification; the full-cohort concordant diagnostic-result rate was 60/64. Our study provides descriptive local performance data, not proof of perfect diagnostic accuracy or evidence that treatment timing, lesion selection, or multidisciplinary planning independently determined success.

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Footnotes

Institutional review board statement: The study was conducted in accordance with the Declaration of Helsinki and was reviewed and approved by the Ethics Committee of Titu Maiorescu University of Bucharest (protocol No. 7/23.03.2026).

Informed consent statement: Written consent for the clinical intervention was obtained as part of routine care.

Conflict-of-interest statement: All the authors report no relevant conflicts of interest for this article.

Data sharing statement: The underlying clinical data are available from the corresponding author at cosmin.moldovan@prof.utm.ro on reasonable request, subject to applicable institutional approval and patient-confidentiality requirements.

STROBE statement: The authors have read the STROBE Statement-checklist of items, and the manuscript was prepared and revised according to the STROBE Statement-checklist of items.

AI contribution statement: Portions of this manuscript were edited using AI tools (GPT-5.6 Sol model) solely for language refinement. The authors carefully reviewed and verified all AI-assisted outputs and take full responsibility for the scientific content of the manuscript.

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Novelty: Grade B, Grade B, Grade C, Grade C

Creativity or innovation: Grade B, Grade B, Grade B, Grade C

Scientific significance: Grade A, Grade B, Grade B, Grade B

P-Reviewer: Chika CE, PhD, Nigeria; Li YC, MD, Associate Chief Physician, China; Meng QY, China **S-Editor:** Liu JH **L-Editor:** A **P-Editor:**

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Figures and Tables

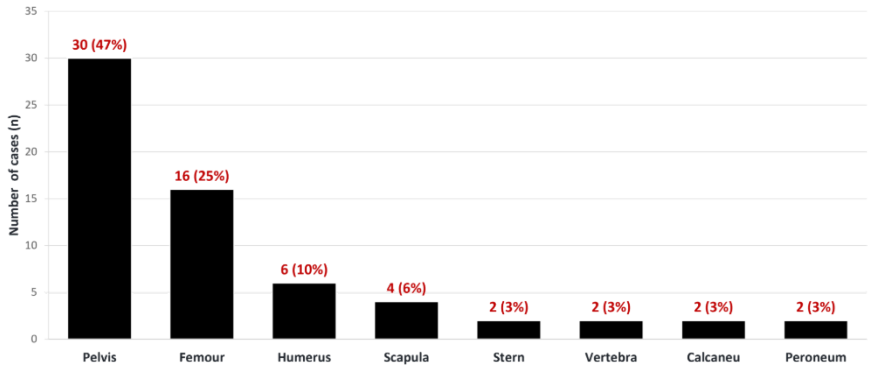


Figure 1 Anatomical distribution of the 64 included biopsy procedures.

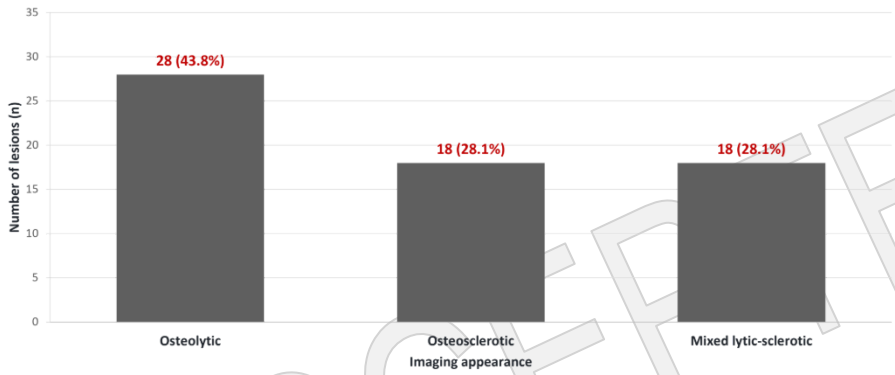


Figure 2 Imaging morphology of the 64 included biopsy procedures: osteolytic, 28/64 (43.8%); osteosclerotic, 18/64 (28.1%); mixed lytic-sclerotic, 18/64 (28.1%). Percentages use all 64 procedures as the denominator.

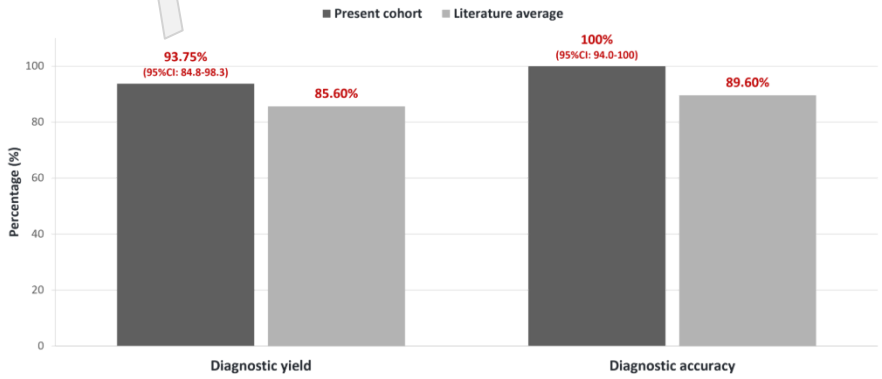


Figure 3 Procedure disposition and diagnostic-verification availability. This flow diagram replaces the unsupported cohort-versus-literature-average chart. NR means not reported, not zero. Counts of 64, 60, and four are taken from the supplied

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manuscript; screening/exclusion totals and unique-patient linkage require source verification (AQ1-AQ3).

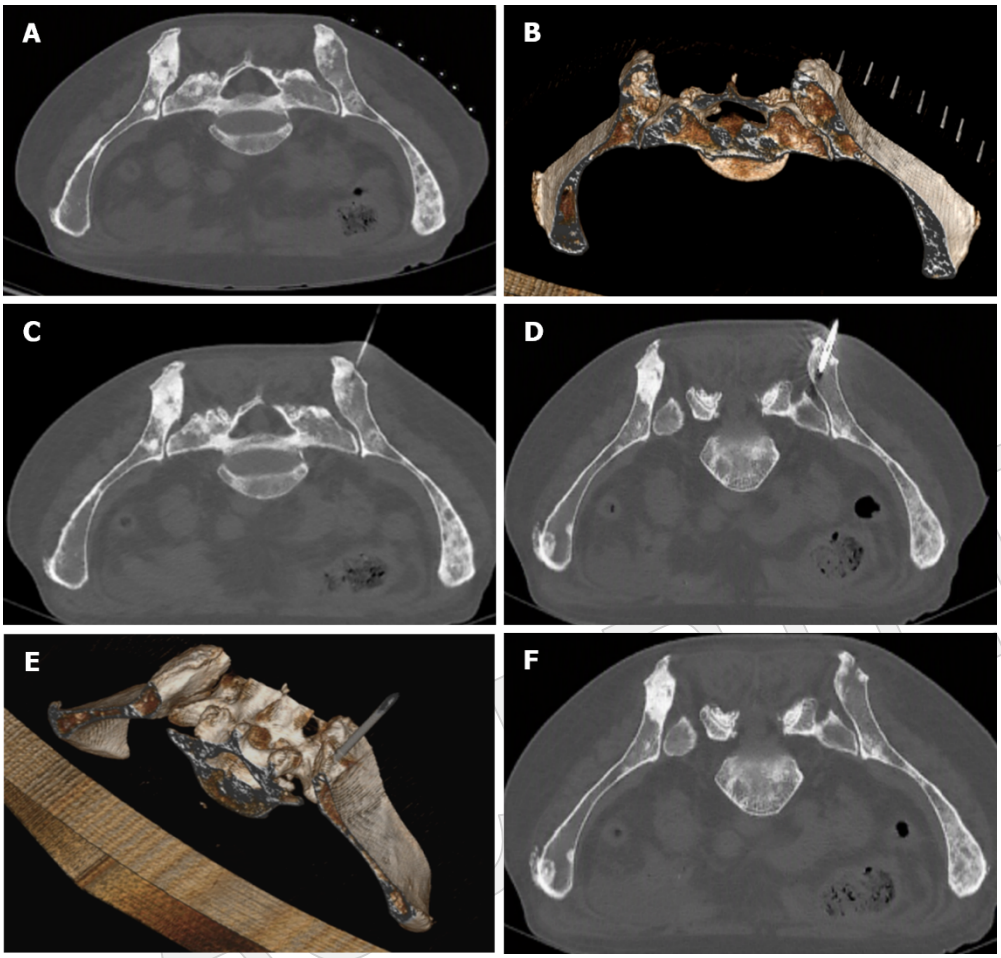


Figure 4 Computed tomography-guided needle biopsy of an osteocondensing lesion. A: Needle computed tomography (CT) scan on the horizontal line; B: Needle scan on the horizontal line with three-dimensional reconstruction; C: Needle guidance under local anesthesia; D: Simple CT scan of the intralesional biopsy needle; E: Three-dimensional CT reconstructed view of the intralesional needle; F: Control scan, confirming the absence of local complications.

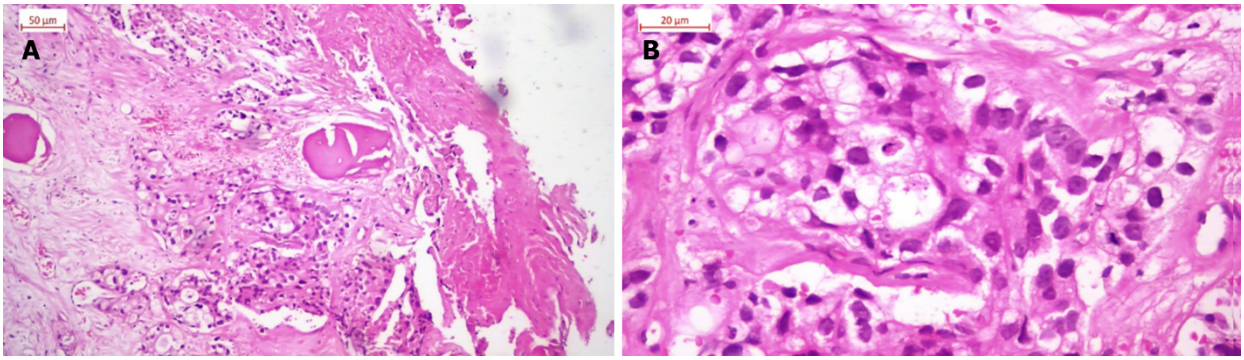
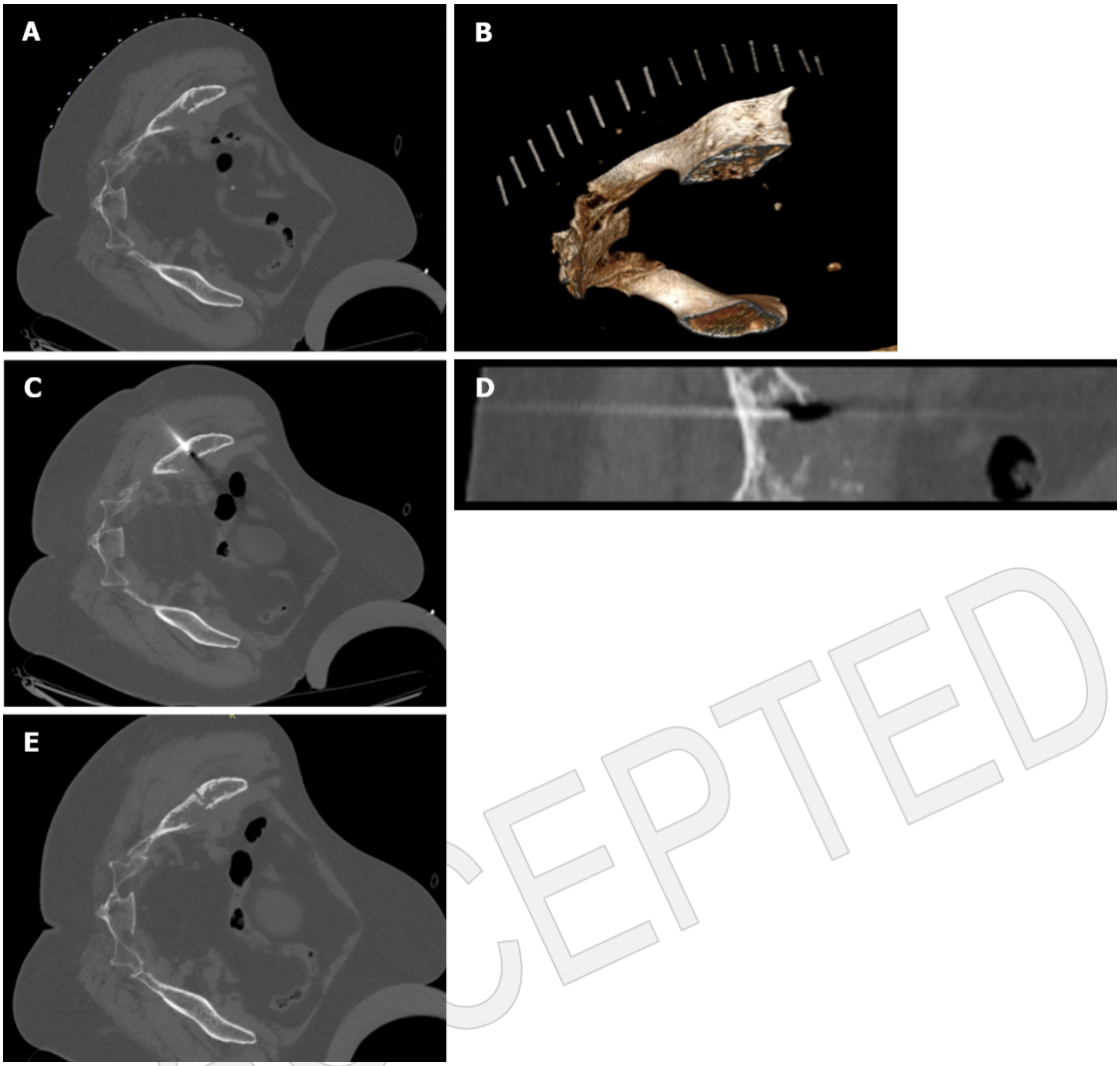


Figure 5 Histopathological appearance of the biopsy specimen in Case 1. Bone tissue showing histological structure altered by malignant epithelial neoplastic proliferation with the histopathological appearance of poorly differentiated adenocarcinoma (G3) with glandular (distinct and cribriform) and solid development pattern. A: Hematoxylin and eosin staining, 100 × optical zoom; B: Hematoxylin and eosin staining, 400 × optical zoom.

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Figure 6 Computed tomography-guided needle biopsy of an osteolytic lesion. A: Needle computed tomography (CT) scan on the horizontal line; B: Needle CT scan on the horizontal line, with three-dimensional reconstruction; C: Intralesional needle biopsy, whole CT scan view; D: Intralesional needle biopsy, zoomed-in view with needle contact; E: Control CT scan showing the complete absence of local complications

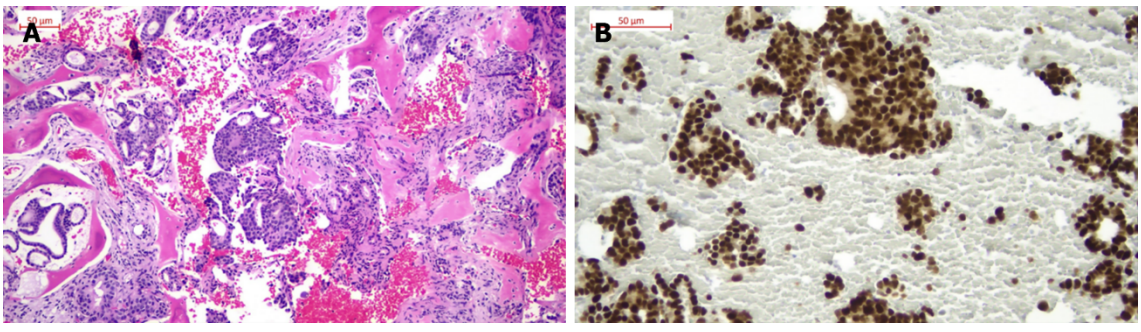


Figure 7 Histopathological and immunohistochemical appearance of the biopsy specimen in Case 2 A: Fragments of trabecular bone tissue (black arrow) in which malignant neoplastic epithelial proliferation is identified, consisting of tubular, glandular and cribriform structures with rare solid structures (clusters, nests) and isolated cells, embedded in a desmoplastic stroma (hematoxylin and eosin staining, 100 × optical zoom); B: Estrogen receptor expression demonstrated by immunohistochemical staining, with intense and diffuse positivity in tumor cell nuclei (200 × optical zoom).

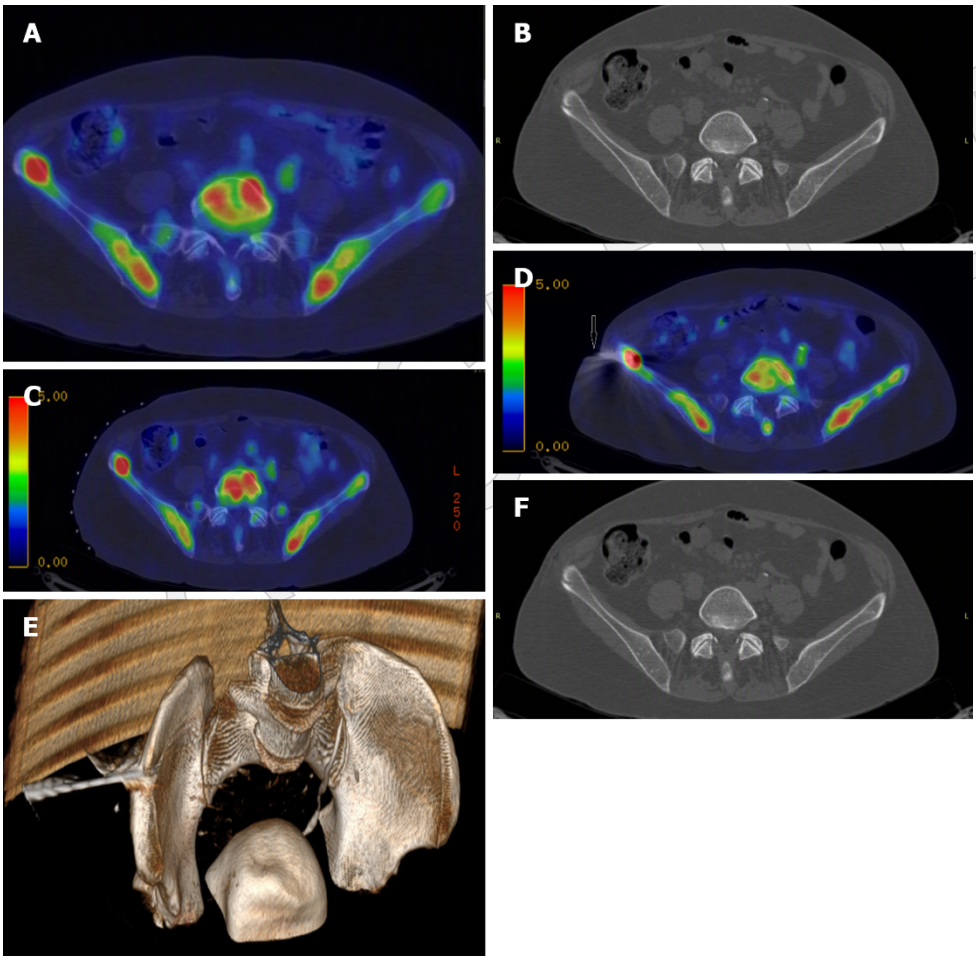


Figure 8 Positron emission tomography/computed tomography-guided needle biopsy of a metabolically active bone lesion. A: Identification of the lesion, visible only on positron emission tomography/computed tomography (PET-CT); B: The lesion is not visible on conventional computed tomography; C: PET-CT scan with pins on the

horizontal line for guidance; D: Intralesional needle biopsy under PET-CT guidance; E: Intralesional PET-CT-guided needle biopsy; F: Control scan, confirming the absence of local complications.

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Table 1 Legacy Society of Interventional Radiology A-F outcome-based complication classification[10]

SIR class	Severity group	Definition
A	Minor complication	No therapy required; no consequence
B	Minor complication	Nominal therapy required; no consequence; may include overnight admission for observation only
C	Major complication	Requires therapy; minor hospitalization for less than 48 hours
D	Major complication	Requires major therapy, unplanned increase in level of care, prolonged hospitalization for more than 48 hours
E	Major complication	Results in permanent adverse sequelae
F	Major complication	Death

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Table 2 Clinical categories and verification descriptions retained from the aggregate source record

Source clinical category	*n*	Reported sites (not exhaustive)	Recorded verification description
Secondary lesions: Prostate carcinoma	10	Iliac wing (4); ischial branch (1). Counts are incomplete	IHC-supported assessment and response to ADT, ADT + docetaxel, or ADT + docetaxel + ARPI. Independence of verification not established

Source clinical category	*n*	Reported sites (not exhaustive)	Recorded verification description
Secondary lesions: Breast carcinoma	10	Iliac wing (2), femoral condyle, iliac body and iliac bone	IHC-supported assessment and response to CDK4/6 inhibition, trastuzumab-pertuzumab, dose-dense docetaxel, or epirubicin + cyclophosphamide, as listed in the source
Secondary lesions: Thyroid, colon, lung and urachal primaries	8	Scapula, femoral metaphysis, trochanter and L5 posterior element	IHC-supported assessment and response to cabozantinib, FOLFIRINOX, pembrolizumab or trastuzumab, as listed in the source
Osteoid osteoma, with radiofrequency ablation	4	Femoral trochanter and humeral condyle	Symptom resolution reported. Confirm actual biopsy histology and the reference-standard rule (AQ1-AQ2)
Benign lesions (enchondroma, cyst, vascular lesion)	8	Femoral condyle/head, humeral head and sternum	Surgical pathology or imaging follow-up; separate counts and dates not reported

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Source clinical category	*n*	Reported sites (not exhaustive)	Recorded verification description
Other primary/recurrent neoplasms: Chondrosarcoma, Ewing sarcoma, recurrent synovial sarcoma, giant cell tumor and multiple myeloma	12	Proximal femur, iliac wing (2), iliac bone, humeral head and scapula	Resection histology; myeloma assessed by IHC with response to D-VTD. Exact route counts not reported. The grouped category requires classification review
Combined discrepancy/non-neoplastic category	8	Peroneal malleolus, calcaneus, femoral condyle and iliac bone	Clinical/imaging follow-up reported; this mixed category is not equivalent to the six diagnosis-correction events
Non-diagnostic procedures	4	Iliac lesions	Final diagnosis/follow-up unavailable in the source description
Total source procedure records	64	Complete site counts pending	Verification methods are not mutually exclusive or individually enumerated

Source-data note: The three metastatic categories sum to 28 records, whereas an earlier narrative stated 59%; no replacement metastatic percentage has been asserted. Individual location counts in the source table were incomplete. These discrepancies require the original procedure-level records, not extrapolation from the older 32-case cohort. IHC: Immunohistochemistry; ADT: Androgen deprivation therapy; ARPI: Androgen receptor pathway inhibitor.

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Table 3 Procedure-level outcomes and two-sided exact 95% confidence intervals

Outcome	Numerator/denominator	Estimate (%)	Exact 95%CI (%)
Diagnostic yield	60/64	93.75	84.8-98.3
Conditional recorded concordance	60/60	100.00	94.0-100.0
Full-cohort concordant diagnostic-result rate	60/64	93.75	84.8-98.3
Non-diagnostic procedures	4/64	6.25	1.7-15.2
Recorded minor complications	2/64	3.13	0.4-10.8
Recorded major complications	0/64	0.00	0.0-5.6
No recorded complication	62/64	96.88	89.2-99.6

The denominator of 60 excludes the four non-diagnostic procedures; the full-cohort endpoint counts them as failures. Final disease status in those four procedures remains unverified. Exact intervals quantify binomial sampling uncertainty and do not resolve verification, incorporation, selection, or surveillance bias. Independence of procedure records requires author confirmation (AQ1).

Table 4 Availability of final verification and follow-up information

Verification item	What the supplied record supports	What remains unresolved

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Verification item	What the supplied record supports	What remains unresolved
All included procedures	64 records	Unique patient count, repeat-procedure handling, and all screened/excluded totals (AQ1)
Diagnostic procedures with recorded final assessment	60/64, all reported concordant	Case-level matching of diagnoses, malignancy/subtype/grade rules, adjudicator identity and blinding (AQ2)
Surgical histopathology	Used for some resected lesions	Exact n/64 and n/60, dates, and independence from the index interpretation (AQ2)
IHC-supported oncologic work-up	Reported for the three metastatic categories (10 + 10 + 8) and for some myeloma cases	Exclusive route counts, source specimen, overlap with surgery/follow-up, and category reconciliation (AQ2/AQ4)
Clinical and imaging follow-up	Used in several source categories; six-month minimum stated previously	Actual duration for each case, median/range, final cutoff, number with ≥ 6 months, and losses to follow-up (AQ3)
Non-diagnostic procedures	4/64; no confirmed final diagnosis/follow-up reported	Subsequent outcome and any repeat procedure; do not reclassify as benign or false-negative without evidence

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Verification item	What the supplied record supports	What remains unresolved
Adverse-event surveillance	Immediate CT assessment; two delayed local fistulas described	Complete surveillance interval and missing follow-up for all 64 (AQ7)
Baseline and procedural variables	Aggregate sex, morphology, anesthesia, guidance categories	Age, lesion size/depth, complete sites, actual cores, treatment exposure, and missingness counts (AQ4-AQ5)

"Not reported" identifies a gap in the supplied summary, not proof that information was never collected. No exclusive reference-standard counts, missingness frequencies, or follow-up durations have been manufactured from the grouped categories. CT: Computed tomography; IHC: Immunohistochemistry.

Table 5 Selected external studies and reviews used for narrative context; outcomes are reported with their original populations and denominators, not pooled

Ref.	Design and population	Verified endpoint/finding	Interpretive boundary
Wang *et al*[16], 2019	Retrospective; 31 patients with bone lymphoma	Diagnosis and subtype established in 26/31 (83.9%). PET/CT findings informed target selection	Selected lymphoma population; not a general bone-tumor accuracy denominator

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Ref.	Design and population	Verified endpoint/finding	Interpretive boundary
Dablan *et al*[17], 2025	Retrospective; 211 patients, 226 biopsies; CT and ultrasound guidance	Adequacy 148/211 patients (70.1%). Higher SUV was associated with adequacy within the PET-imaged subset	70.1% uses patients, not 226 procedures. Neither CT-only nor an external control group
Filippiadis *et al*[18], 2018	Narrative review of biopsy of bone metastases	Technical guidance on indication, targeting, specimen acquisition, and safety	No original study denominator; not an independent cohort or pooled performance estimate
Maciel *et al*[19], 2014	Retrospective; 186 procedures	Adequate material in 183/186 (98.4%); three complications, including fracture, functional paresthesia, and needle breakage requiring removal	Specimen adequacy is not verified diagnostic accuracy; complications were not all trivial

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Ref.	Design and population	Verified endpoint/finding	Interpretive boundary
Ferreira *et al*[20], 2023	Systematic review/meta-analysis; 29 studies	Pooled CNB yield 91.4%, accuracy 92.7%; FNA yield 88.5%, accuracy 82.5%; combined yield 96.5%	Secondary evidence, not a separate primary cohort. Do not pool again with potentially overlapping primary reports
Li *et al*[21], 2014	Retrospective; 162 biopsies in 155 patients	Yield 81.5%; lytic 89.9% *vs* sclerotic 48.5%; lesions ≥ 3 cm 89.2% *vs* smaller lesions 73.4%	Associations belong to that study; size threshold is ≥ 3 cm. Repeated procedures may differ from unique patients
Hwang *et al*[22], 2011	Retrospective; 800 CT-guided bone biopsies	Initial results: 69% diagnostic, 31% indeterminate. CT features related to diagnostic/indeterminate outcomes were evaluated	Initial yield is distinct from subsequent final diagnosis. This is the CT-feature study, not Hoang *et al*[26]
Suh and Yun[23], 2019	Meta-analysis; 15 studies of sclerotic lesions	Pooled yield 74% (95%CI: 62%-84%; 969 lesions); accuracy 87% (95%CI: 77%-93%; 242 lesions)	Different denominators contribute to yield and accuracy. Restricted to sclerotic lesions

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Ref.	Design and population	Verified endpoint/finding	Interpretive boundary
Toffart *et al*[24], 2020	Retrospective; 50 lytic lesions in patients suspected of lung cancer	Pathological diagnosis 50/50 (100%); molecular-testing success 35/37 (94.6%)	Molecular success is a separate endpoint among tested specimens, not overall biopsy accuracy
Monfardini *et al*[25], 2014	Retrospective; 308 procedures in 286 cancer patients	Adequacy 290/308 (94.1%); sensitivity 96.7% under the study's diagnostic/follow-up definition	Adequacy and sensitivity are not interchangeable; final verification differed from the current cohort
Hoang *et al*[26], 2019	Retrospective; 115 patients with suspected osteomyelitis	Positive cultures in 24/115 (20.9%); treatment changed in 10 of the 24 culture-positive cases	Microbiological yield, not oncologic histopathological accuracy; not specific to bone tuberculosis

The table is a non-systematic selection of clinical cohorts and secondary evidence. Published estimates retain study-specific definitions and are not combined into a "literature average." Bibliographic years refer to the issue publication year (Ferreira *et al*[20], 2023; Toffart *et al*[24], 2020). CNB: Core-needle biopsy; FNA: Fine-needle aspiration; SUV: Standardized uptake value; PET/CT: Positron emission tomography/computed tomography.

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